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★ Member-only story

Finally Tflite Model Maker can work in Colab!

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How do we solve the perennial Tflite Model Maker issue? A lot of researchers was stuck when Google Colab updated to version 3.10 and yet Tflite-Model-Maker still remains at ver 3.9.

Many researchers have tried to downgrade the Colab to version 3.9 using this command `!sudo apt-get install python3.9`, but nothing works!

The solution: we need to run Google Colab at version 3.9 using virtual environment. We also need to run the training commands as a python script, not directly at the cells. Let's see how does this.

1. Get Miniconda

Get the Miniconda using the commands below

```
!wget https://repo.anaconda.com/miniconda/Miniconda3-py39_23.3.1-0-Linux-x86_64.sh
```

```
!chmod +x Miniconda3-py39_23.3.1-0-Linux-x86_64.sh
```

```
!./Miniconda3-py39_23.3.1-0-Linux-x86_64.sh -b -f -p /usr/local
```

```
!conda update conda
```

2. Create conda with python version 3.9

Create a dedicated space (environment) called `myenv` and installing Python 3.9 into it.

```
import sys
```

```
sys.path.append('/usr/local/lib/python3.9/site-packages')
```

```
!conda create -n myenv python=3.9
```

3. Install Tflite-model-maker in Conda env

Virtual Environments help us manage dependencies and ensure consistency.

```
%%shell
```

```
eval "$(conda shell.bash hook)"
```

```
conda activate myenv
```

```
pip install tflite-model-maker
```

4. Install ipykernel to allow python ver3.9 in colab

`ipykernel` lets us create a kernel for the `myenv` environment, allowing you to use its specific Python and packages within a Colab

notebook.

```
%%shell
```

```
eval "$(conda shell.bash hook)"
```

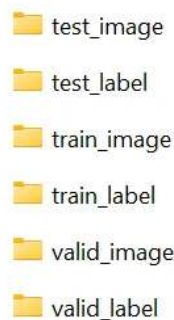
```
conda activate myenv
```

```
pip install ipykernel
```

```
pip install numpy==1.23.4
```

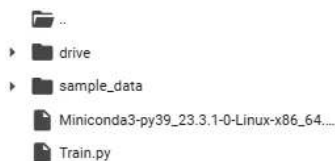
5. Test out with a sample dataset (link below) from Roboflow.

It is always good to test out a sample dataset before running our custom dataset. We need to manually separate into the annotation and images file — in total 6 folders



6. Upload the train.py script

Upload the train.py (link below) to content. Remember to set the path and label_map correctly.



```
label_map = {1: 'player', 2: 'goalkeeper', 3: 'referee', 4: 'ball'}

train_images_dir = '/content/dataset/train_image'
train_annotations_dir = '/content/dataset/train_label'
val_images_dir = '/content/dataset/valid_image'
val_annotations_dir = '/content/dataset/valid_label'
test_images_dir = '/content/dataset/test_image'
test_annotations_dir = '/content/dataset/test_label'
```

7. Finally train the model

Train the model by running this command

```
%%shell
```

```
eval "$(conda shell.bash hook)"
```

```
conda activate myenv
```

```
python Train.py
```

```
Epoch 2/50
29/29 [=====] - ETA: 0s - det_loss: 1.1899 - cls_loss: 0.5107 - box_loss: 0.0136 - reg_
29/29 [=====] - 160s 5s/step - det_loss: 1.1848 - cls_loss: 0.5085 - box_loss: 0.0135 -
```

8. Set hyperparameters in the object detector

Select EfficientDet_lite2 with 100 epochs and batchsize 16 as a start. Do not run on batchsize 32 or 64 as you will get error due to the large memory required.

To speed up, please consider using the smallest model (efficientdet-lite0) and batchsize 16.

```
spec = model_spec.get('efficientdet_lite2')

model = object_detector.create(train_data=train_data,
                              model_spec=spec,
                              validation_data=validation_data,
                              epochs=100,
                              batch_size=16,
                              train_whole_model=True)
```

9. Finally copy the output .tflite and labels to test at Raspberry Pi.

TFLITE_FILENAME = 'trained_model.tflite'

LABELS_FILENAME = 'model-labels.txt'

Reference

1. videoguide:<https://youtu.be/Enysj0IVEno>
2. script:<https://github.com/elvenkim1/tflite-model-maker/blob/main/Train.py>
3. 3) dataset:<https://universe.roboflow.com/roboflow-jvuqo/football-players-detection-3zvbc>



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Written by Elven Kim

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I am a researcher in the field of Robotics, Computer Vision, Artificial Intelligence and Industry Automation.

Responses (1)



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What are your thoughts?



Florian Roscheck
Jul 27, 2025 (edited)

...

Thank you for your guide, @Elven Kim, it has helped me get started with setting up tflite model maker on Google Colab. However, this solution cannot take advantage of Colab's GPU.

I found a way to leverage GPUs for training with tflite model maker... [more](#)



1 reply

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